

Assignment – Module 2

Linear Regression & Logistic Regression

Q1. Scatter Diagram & Model Representation

- a) Draw and explain a scatter diagram for a sample dataset.
- b) Explain the model representation for a single variable in linear regression.

Q2. Cost Function and Least Squares

- a) Derive the Least Squares method for Linear Regression.
- b) Explain the concept of Cost Function and its importance.

Q3. Gradient Descent Method

- a) Explain Gradient Descent with mathematical formulation.
- b) Differentiate between Batch Gradient Descent and Stochastic Gradient Descent.
- c) Discuss convergence issues and how to select a proper learning rate.

Q4. Properties and Assumptions of Linear Regression

- a) Explain the assumptions made in Linear Regression.
- b) Discuss properties of regression line.
- c) Explain how model performance is measured using R^2 .

Q5. Multiple Linear Regression & Regularization

- a) Explain multivariable model representation and cost function.
- b) Describe Overfitting and Underfitting in regression.
- c) Explain Bias-Variance trade-off and the role of Regularization (L1/L2).

Q6. Logistic Regression

- a) Explain why Logistic Regression is used for classification problems.
- b) Derive the Sigmoid Function and explain its properties.
- c) Discuss Odds of an Event, Decision Boundary, and Maximum Likelihood function.

Q7. Logistic vs Linear Regression

- a) Compare Linear Regression and Logistic Regression in terms of assumptions, applications, and limitations.
- b) Explain Pros and Cons of Logistic Regression.

Q8. Evaluation Metrics

- a) Define Confusion Matrix.
- b) Explain Precision, Recall, and Accuracy with suitable formulas and examples.
- c) Discuss how these metrics help measure classification performance.